

Document Representations

CM52065

Where are we?

	Lecture A	Lecture B	Lab
Week 1	Introduction to NLP	Language Preprocessing	Genre Classification
Week 2	Document Representations	Handling Sparsity	Sentiment Analysis
Week 3	Static Word Embeddings	Contextual Word Embeddings	Embedding Vis.
Week 4	Information Extraction	Syntax Parsing	Relationship Extr.
Week 5	Classical Sequence Models	Neural Sequence Models	Text Generation

Learning Objectives

Illustrate BoW, BoN and TF-IDF representations.

Compare and contrast document representations.

Explain how to mitigate the out-of vocabulary problem.

Document Representations: One-hot encoding

- Last week we used one-hot encoding to represent documents as feature vectors.

Document	dogs	love	other	i	and	cats	do	not	to	Feature Vector
dogs love other dogs	1	1	1	0	0	0	0	0	0	[111000000]
i love dogs and cats	1	1	0	1	1	1	0	0	0	[110111000]
cats do not love cats	0	1	0	0	0	1	1	1	0	[010001110]
i love to love.	0	1	0	1	0	0	0	0	1	[010100001]

Question

Using this representation can you tell me which document is most relevant to the topic of dogs?

Document Representations: One-hot encoding

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i love dogs and cats	1	1	0	1	1	1	0	0	0	[110111000]
cats do not love cats	0	1	0	0	0	1	1	1	0	[010001110]
i love to love.	0	1	0	1	0	0	0	0	1	[010100001]

- This type of text encoding is sometimes referred to as binary bag of words.

Document Representations: Bag of Words

- Binary bag of words track only whether a word has occurred or not.
- Bag of words (BoW) tracks how many times a word appears in a document.

Document	dogs	love	other	i	and	cats	do	not	to	Feature Vector
dogs love other dogs	1	1	1	0	0	0	0	0	0	[111000000]
i love dogs and cats	1	1	0	1	1	1	0	0	0	[110111000]
cats do not love cats	0	1	0	0	0	1	1	1	0	[010001110]
i love to love.	0	1	0	1	0	0	0	0	1	[010100001]

Document Representations: Bag of Words

- Binary bag of words track only whether a word has occurred or not.
- Bag of words (BoW) tracks how many times a word appears in a document.

Document	dogs	love	other	i	and	cats	do	not	to	Feature Vector
dogs love other dogs	2	1	1	0	0	0	0	0	0	[211000000]
i love dogs and cats	1	1	0	1	1	1	0	0	0	[110111000]
cats do not love cats	0	1	0	0	0	1	1	1	0	[010001110]
i love to love.	0	2	0	1	0	0	0	0	1	[020100001]

Document Representations: Bag of Words

- Binary bag of words track only whether a word has occurred or not.
- Bag of words (BoW) tracks how many times a word appears in a document.
- Let's take another example, we are looking for positive reviews of a given movie.

Document	...	great	...
great movie	...	1	...
A great, great, great movie	...	3	...

Question

Which document is more positive about the movie?

Document Representations: Bag of Words

- Binary bag of words track only whether a word has occurred or not.
- Bag of words (BoW) tracks how many times a word appears in a document.
- Let's take another example, we are looking for positive reviews of a given movie.

Document	...	great	...
great movie	...	1	...
A great, great, great movie	...	3	...

- Just because a document is longer and contains more tokens, it may not be better!
 - Consider two CVs, one is 6 pages long, another is 1 page long, imagine the BoW feature vectors.

Document Representations: Bag of Words

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- Bag of words (BoW) tracks how many times a word appears in a document.
- BoW does not consider document length.

Document Representations: Bag of Words

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- Bag of words (BoW) tracks how many times a word appears in a document.
- BoW does not consider document length.

Question

Complete the table below using BoW.

Bonus: What do you learn about the BoW representation?

Document	the	cat	sat	on	the	dog	Feature Vector
The cat sat on the dog	1	1	1	1	1	1	[111111]
The dog sat on the cat	1	1	1	1	1	1	[111111]

Document Representations: Bag of Words

- Binary bag of words track only whether a word has occurred or not.
- Bag of words (BoW) tracks how many times a word appears in a document.
- BoW does not consider document length.
- BoW does not consider word order.

Document Representations: Bag of Words

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- Bag of words (BoW) tracks how many times a word appears in a document.
- BoW does not consider document length.
- BoW does not consider word order.

**Let's look at resolving
these limitations**

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

the cat sat on the dog

Unigrams

the

on

cat

dog

sat

Bigrams

the cat

on the

cat sat

the dog

sat on

Trigrams

the cat sat

on the dog

cat sat on

sat on the

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

Document						
the cat sat on the dog						
the dog sat on the cat						

Question

Complete the table below using Bag of 3-Grams (Trigrams).

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

Document	the cat sat	cat sat on	sat on the	on the dog	the dog sat	on the cat
the cat sat on the dog	1	1	1	1	0	0
the dog sat on the cat	0	0	1	0	1	1

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

Document	Bag of Words	Bag of 3-Grams
the cat sat on the dog	[1, 1, 1, 1, 1, 1]	[1, 1, 1, 1, 0, 0]
the dog sat on the cat	[1, 1, 1, 1, 1, 1]	[0, 0, 1, 0, 1, 1]

- We are now able to distinguish between the two documents.

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

Document	Bag of Words	Bag of 2-Grams	Bag of 3-Grams
the cat sat on the dog	[1, 1, 1, 1, 1, 1]		[1, 1, 1, 1, 0, 0]
the dog sat on the cat	[1, 1, 1, 1, 1, 1]		[0, 0, 1, 0, 1, 1]

- We are now able to distinguish between the two documents.

Question

Fill in the blanks for 2-Grams.

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.

Document	Bag of Words	Bag of 2-Grams	Bag of 3-Grams
the cat sat on the dog	[1, 1, 1, 1, 1, 1]	[1, 1, 1, 1, 1, 1]	[1, 1, 1, 1, 0, 0]
the dog sat on the cat	[1, 1, 1, 1, 1, 1]	[1, 1, 1, 1, 1, 1]	[0, 0, 1, 0, 1, 1]

- We are now able to distinguish between the two documents.
 - But we're depending on selecting the right N-Grams.

Document Representations: Bag of N-Grams

- To overcome the limitation of word order, we introduce N-Grams.
- An n-gram is a sequence of tokens from text.
- We are now able to distinguish between the two documents.
 - But we're depending on selecting the right N-Grams.
- Okay, so we have mitigated the ordering of tokens using N-Grams with the limitation of needing to select N.

Document Representations: TF-IDF

- BoW and BoN-Grams treat all tokens/N-grams equally but these approaches do not consider document length.
- TF-IDF is a representation which determines how important each token/N-Gram is to the document using two components:
 - How often a term appears in a document.
 - How often a term appears in the corpus.
- These considerations reduce the relevance of common words and increase the relevance of rare words.
 - This is incredibly important if you aren't removing stop words!

Document Representations: TF-IDF

- To calculate the TF-IDF of a document we use the following equation:

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$

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Frequency of term t in document d .

Rarity of ter

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$$TF(\text{'sun'}, \text{'the sun rose high in the sky'}) = 1 / 7 = 0.143$$

$$TF(\text{'the'}, \text{'the sun is the best'}) = 1 / 5 = 0.200$$

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$$TF(t, d) = \frac{\text{Number of times } t \text{ occurs in } d}{\text{Number of terms in } d}$$

This shows us how often a word appeared in a document.

$$TF(\text{'sun'}, \text{'the sun rose high in the sky'}) = 1 / 7 = 0.143$$

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Document Representations: TF-IDF

- To calculate the TF-IDF of a document we use the following equation:

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$

$$IDF(t) = \log_e \left(\frac{\text{Number of documents}}{\text{Number of documents containing } t} \right)$$

Documents:

['the sun rose high in the sky', 'the sun is the best', 'the moon is the best']

$$IDF(sun) = \log_e \left(\frac{3}{2} \right) IDF(moon) = \log_e \left(\frac{3}{1} \right) IDF(the) = \log_e \left(\frac{3}{3} \right)$$

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**This shows us the rarity
of a given word in the
corpus!**

Documents:

['the sun rose high in the sky', 'the sun is the best', 'the moon is the best']

$$IDF(sun) = 0.41 \quad IDF(moon) = 1.10 \quad IDF(the) = 0$$

Document Representations: TF-IDF

- To calculate the TF-IDF of a document we use the following equation:

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$

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Preprocesses Documents

sun rose high sky

sun best

moon best rose high

Helper	sun	rose	high	sky	best	moon
IDF(t)	0.41	0.41	0.41	1.10		
TF(t, 0)	0.25	0.25	0.25	0.25	0	0
TF(t, 1)	0.5	0	0	0	0.5	0
TF(t, 2)						

Question

Fill in the blanks. Do you notice any patterns?

Document Representations: TF-IDF

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Helper	sun	rose	high	sky	best	moon	Preprocesses Documents
IDF(t)	0.41	0.41	0.41	1.10	0.41	1.10	sun rose high sky
TF(t, 0)	0.25	0.25	0.25	0.25	0	0	sun best
TF(t, 1)	0.5	0	0	0	0.5	0	moon best rose high
TF(t, 2)	0	0.25	0.25	0	0.25	0.25	
TF-IDF(t, 0)							
TF-IDF(t, 1)							
TF-IDF(t, 2)							

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TF(t, 0)	0.25	0.25	0.25	0.25	0	0
TF(t, 1)	0.5	0	0	0	0.5	0
TF(t, 2)	0	0.25	0.25	0	0.25	0.25
TF-IDF(t, 0)	0.103	0.103	0.103	0.275	0	0
TF-IDF(t, 1)	0.205	0	0	0	0.205	0
TF-IDF(t, 2)	0	0.103	0.103	0	0.103	0.275

Preprocesses Documents
sun rose high sky
sun best
moon best rose high

Question

Which words are most (or least) important and why?

Document Representations: Comparison

Bag of Words

Limited technical skills required to implement and interpret

Computationally Efficient

Loses order and context

Sensitive to common tokens

Beneficial use-cases:

- Text Classification
- Topic Modelling
- Small datasets
- Finding a quick baseline

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Bag of N-Grams

Captures context (e.g. 'not ok').

Domain-specific expressions (e.g. machine learning model)

Vocabulary size scales rapidly

Requires more memory and compute

Beneficial use-cases:

- Word order matters
- Large dataset available
- Named Entity Recog.
- Short documents

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TF-IDF

Can be used in conjunction with BoW or BoNG

Reduces noise from stop words

Ignores semantics

Sparse and high-dimensional feature vectors

Beneficial use-cases:

- Keyword identification
- Text classification
- Search / Querying
- Clustering

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Document Representations: OOV problem

- All three of these representations have one problem we saw in the lab last week.
 - What happens when we perform inference on a sentence which contains tokens not in the vocabulary?

Bag of Words

Bag of N-Grams

TF-IDF

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Bag of Words

Processed Document:

- [dogs, singing, tonight]

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Bag of Words

Processed Documents:

- [dogs, singing, night]
- [cats, bark, sun]



Vocabulary:

- sky
- night
- moon
- sun

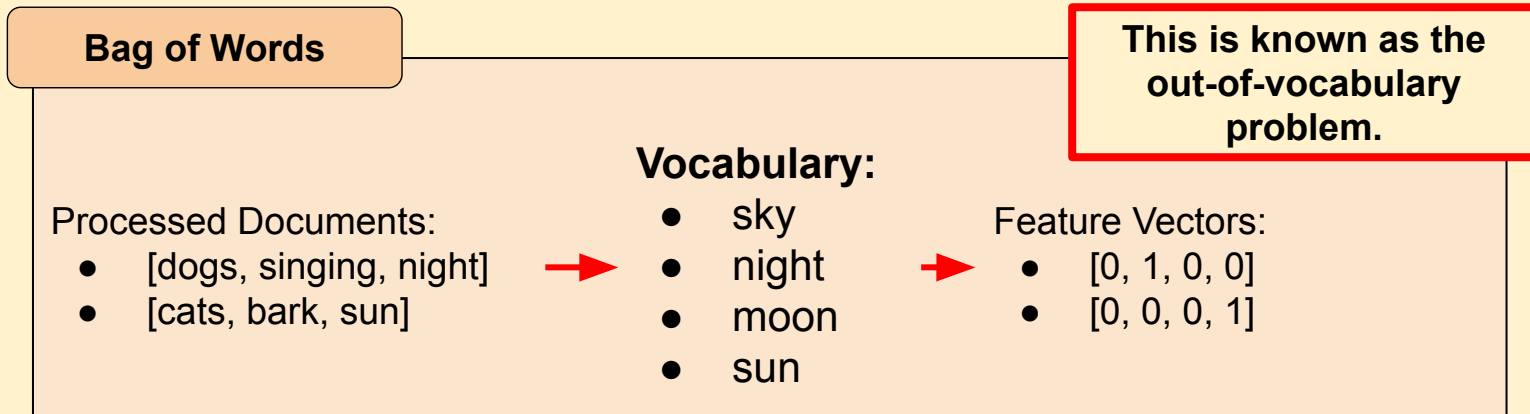


Feature Vectors:

- [0, 1, 0, 0]
- [0, 0, 0, 1]

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Document Representations: OOV problem

- The OOV problem occurs when the trained vocabulary doesn't match the full domain of expected tokens.
 - This is partially mitigated through preprocessing.

Removal of
stop words

Tokenisation

Lemmatisation

Stemming

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- Two other popular methods:
 - Character n-grams
 - <UNK> tokens

Document Representations: OOV problem

<UNK> Tokens

- Use a '<UNK>' token for:
 - Unknown tokens
 - Infrequent tokens (e.g. low IDF)

[cats, bark, sun]

Vocabulary:

- sky
- night
- moon
- sun
- <UNK>

→ [0, 0, 0, 1, 2]

- Reduces dimensionality of large vocabularies.
- Loses some information.

Document Representations: OOV problem

<UNK> Tokens

- Use a '<UNK>' token for:
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Vocabulary:

- sky
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- <UNK>

→ [0, 0, 0, 1, 2]

- Reduces dimensionality of large vocabularies.
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Character n-grams

- Use character-level n-grams instead of standard word-level n-grams.

[dogs, dont, eat, bugs]



[do, gs, nt, ea, t, bu, gs]

[languages]



[lan, gua, ges,]

- Handles misspellings
- Loss of semantic meaning
- Very large feature vectors

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- Two other popular methods:
 - Character n-grams
 - <UNK> tokens
- Other options: **Hashing trick**, combining word and character n-grams.



Use-Case: Document Search

- mobius technologies and services is working to provide a client with a document search facility.
- Specifically they aim to:

To present rich and accurate analytics and research intelligence, the company had to delve into a treasure trove of information seen in a huge volume of industry reports and presentations. A robust, scalable, and easily accessible web-based document search management system became the need of the hour to smoothen collation of data by the researchers.



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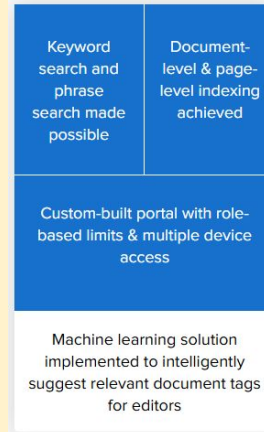
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- The solution overview:





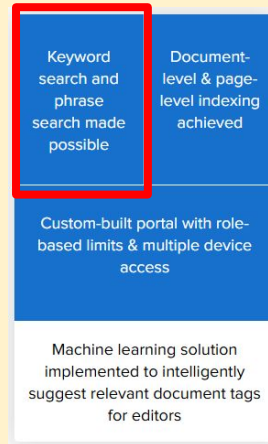
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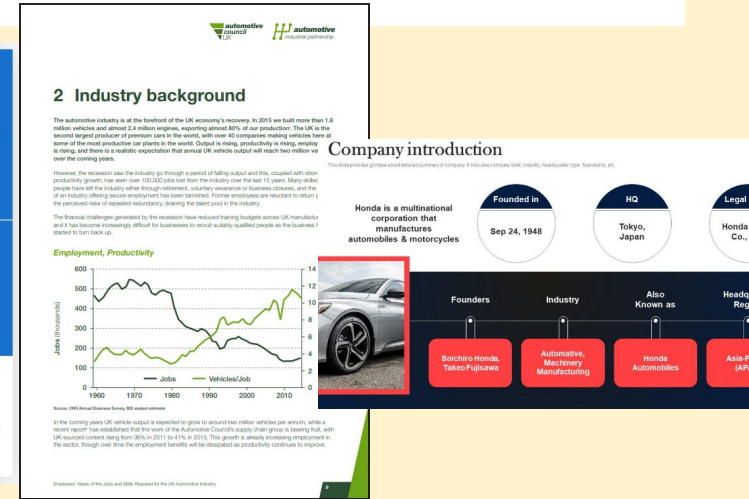
To present rich and accurate analytics and research intelligence, the company had to delve into a treasure trove of information seen in a huge volume of industry reports and presentations. A robust, scalable, and easily accessible web-based document search management system became the need of the hour to smoothen collation of data by the researchers.

Keyword search and phrase search made possible

Document-level & page-level indexing achieved

Custom-built portal with role-based limits & multiple device access

Machine learning solution implemented to intelligently suggest relevant document tags for editors



Learning Objectives

Illustrate BoW, BoN and TF-IDF representations.

Compare and contrast document representations.

Explain how to mitigate the out-of vocabulary problem.

Feedback

- We have 10 weeks together, let me know how it is going!
- This is my first time running this course, so I am keen to hear your thoughts.
- It's never too early or too late to provide feedback.

<https://forms.gle/9WsY57cZKSkLQz8k6>



CM52065 Natural Language Processing

Hi folks,

Thanks for taking the time to complete this very short feedback form, this is all anonymous and will be checked regularly!

Thanks,
Andy

* Indicates required question

Feedback *

Your answer

Submit Clear form